CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: December 2024 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Discuss Primitive Data Types in Java.

Java Data Types:

- **Integer Types:** `byte` (1 byte), `short` (2 bytes), `int` (4 bytes), `long` (8 bytes).
- **Floating-Point:** `float` (4 bytes, IEEE 754), `double` (8 bytes).
- **Character:** `char` (2 bytes, 16-bit Unicode character).
- **Boolean:** `boolean` (represents true or false).

Q2 (3M) Explain TCP/IP Server Socket in Java.

Explanation: Java provides the `java.net.ServerSocket` class to implement a TCP server that listens for incoming connection requests on a designated port.

```
ServerSocket server = new ServerSocket(8080);
Socket client = server.accept(); // Blocks until connection arrives
```

Q3 (3M) What are Packages? How do you call a package in Java?

Definition: A **Package** is a namespace mechanism that organizes related classes and interfaces into modular directory structures, preventing naming conflicts and providing access protection.
Calling a Package: Uses the `import` keyword:

```
import java.util.ArrayList; // Imports single class
import java.io.*;           // Imports all classes in package
```

Q4 (3M) Explain the Anchor and Table tags in HTML.

HTML Tags:

- `<a href="url">Link Text</a>` : Creates a hyperlink to navigate between web pages or download files.
- `<table><tr><th>...</th><td>...</td></tr></table>` : Renders structured tabular data using rows (`<tr>`), headers (`<th>`), and data cells (`<td>`).

Q5 (3M) Briefly explain Internet Concepts: URL, HTTP, and Web Server.

Concepts:

- **URL (Uniform Resource Locator):** Global address of a resource on the web (e.g., `https://www.example.com/index.html`).
- **HTTP:** HyperText Transfer Protocol, a stateless client-server protocol running over TCP port 80/443 for web communication.
- **Web Server:** Hardware/software that serves web resources to clients (Apache, Nginx, Tomcat).

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Explain Applet Life Cycle. Write a program to demonstrate a simple Java applet to display an image. Compare Applets over HTML.**

Applet Image Display Code:

```
import java.applet.*;
import java.awt.*;

/* <applet code="ImageApplet.class" width="400" height="300"></applet> */
public class ImageApplet extends Applet {
    Image img;
    public void init() {
        img = getImage(getDocumentBase(), "college_logo.png");
    }
    public void paint(Graphics g) {
        g.drawImage(img, 20, 20, this);
        g.drawString("CCSU BCA V Semester", 20, 260);
    }
}
```

Comparison over HTML: Applets provide dynamic client-side computation, graphical interactivity, and local state management that static HTML cannot natively provide.

Q7 (7.5M) **Explain TCP/IP Client Socket. Write constructors and methods used to create client sockets.**

Client Socket (java.net.Socket):

- **Constructors:**
  - `Socket(String host, int port)` — Establishes TCP stream connection.
  - `Socket(InetAddress address, int port)`
- **Methods:**
  - `getInputStream()` — Returns InputStream for reading server responses.
  - `getOutputStream()` — Returns OutputStream for sending data.
  - `close()` — Gracefully closes the TCP connection.

Q8 (7.5M) **Explain the concepts of Inheritance in Java and its various types.**

Inheritance: The mechanism where a subclass derives attributes and behaviors from a superclass using `extends`.
Types in Java:

1. **Single Inheritance:** Class B extends Class A.
2. **Multilevel Inheritance:** Class C extends Class B, which extends Class A.
3. **Hierarchical Inheritance:** Class B and Class C both extend Class A.
4. **Multiple Inheritance:** Achieved through **Interfaces** (Java does not support multiple class inheritance to avoid the Diamond Problem).

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **What is JDBC? Write a complete JDBC program to insert and display records of employees using PreparedStatement.**

Complete PreparedStatement JDBC Program:

```
import java.sql.*;

public class EmployeeJDBC {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/company_db";
        String user = "root";
        String password = "password";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection conn = DriverManager.getConnection(url, user, password);

            // 1. Insert record using PreparedStatement
            String insertSql = "INSERT INTO Employee(EmpID, Name, Salary) VALUES(?, ?, ?)";
            PreparedStatement psInsert = conn.prepareStatement(insertSql);
            psInsert.setInt(1, 101);
            psInsert.setString(2, "Rahul Sharma");
            psInsert.setDouble(3, 55000.00);
            int rows = psInsert.executeUpdate();
            System.out.println("Inserted rows: " + rows);

            // 2. Display all employee records
            String selectSql = "SELECT EmpID, Name, Salary FROM Employee";
            Statement stmt = conn.createStatement();
            ResultSet rs = stmt.executeQuery(selectSql);

            System.out.println("\n--- EMPLOYEE RECORDS ---");
            while(rs.next()) {
                System.out.println(rs.getInt("EmpID") + "\t" +
                                   rs.getString("Name") + "\t" +
                                   rs.getDouble("Salary"));
            }
            conn.close();
        } catch(Exception e) {
            e.printStackTrace();
        }
    }
}
```

Q10 (15M) **What are Packages? Explain Java API Packages. List any five methods of the String class with examples.**

String Class Methods:

1. `length()` : Returns character count → `"Hello".length() == 5` .
2. `charAt(int index)` : Returns character at index → `"Hello".charAt(1) == 'e'` .
3. `substring(int begin, int end)` : Extracts substring → `"Java".substring(0, 2) == "Ja"` .
4. `toUpperCase()` : Converts to uppercase → `"bca".toUpperCase() == "BCA"` .
5. `equals(Object another)` : Compares string contents → `"Java".equals("Java") == true` .

Q11 (15M) **Explain Servlets with their Life Cycle. How is it different from the Life Cycle of JSP?**

Comparison:

- **Servlet:** Java class compiled into bytecode; container directly executes `init()`, `service()`, and `destroy()`.
- **JSP:** Text-based document with HTML + embedded Java code; container first translates JSP into a temporary Java Servlet source file, compiles it to bytecode, and invokes `_jspInit()`, `_jspService()`, and `_jspDestroy()`.

Q12 (15M) **Discuss JSP in detail. What are JSP Directives? Explain various types of directives with examples.**

JSP Directives (`<%@ directive attribute="value" %>`):

1. **page Directive:** Controls page-level settings: `<%@ page import="java.util.*, java.sql.*" contentType="text/html" errorPage="error.jsp" %>`
2. **include Directive:** Inserts static files at translation time: `<%@ include file="header.jsp" %>`
3. **taglib Directive:** Imports custom tag libraries (JSTL): `<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>`

Q13 (15M) **What are Layout Managers? Explain five Layout Managers supported by AWT/Swing.**

Five Layout Managers:

1. **FlowLayout:** Arranges components row-by-row like words in a paragraph.
2. **BorderLayout:** Positions in North, South, East, West, Center regions.
3. **GridLayout:** Rectangular grid of equal-sized cells.
4. **CardLayout:** Stacks components like a deck of cards (one visible at a time).
5. **GridBagLayout:** Most flexible layout manager using `GridBagConstraints` for variable cell spans and weights.
